



Well Construction and Completion Design

Well control during completions, workover, and interventions requires a broad understanding of how the well is constructed, familiarity with different completion designs, and a good understanding of wellhead and downhole completion equipment. Armed with this knowledge, those planning and executing completion and intervention activities will be able to identify potential leak paths, and maintain an increased level of vigilance during operations where the risk of loss of containment is increased. A good understanding of the well construction process and completion architecture will also enable rig site personnel to respond more effectively in the event of a well control incident.

This chapter of the book describes the well construction process (drilling and casing the well) before going on to discuss completions and completion design.



2.1 WELL CONSTRUCTION

Nearly all wells are constructed using a series of casing strings, where each string reduces (tapers down) in diameter as the well gets deeper. Each casing is cemented in place, bonding it to the borehole. A properly cemented casing string will prevent formation pressure migrating to the surface, provide isolation between layers in the formation, support the borehole, and strengthen the casing. A series of tapered and cemented pipes is a construction concept that will be instantly recognized by anyone who has worked in the upstream side of the oil industry for any time.

Casing design is normally the remit of the drilling engineer, who will strive to find a design that allows the well to be drilled safely and economically, whilst at the same time resisting the many forces that are a necessary part of the drilling process. Casing should be designed to maintain integrity throughout the life of the well.

Drilling the well normally starts with the placement of the surface conductor. For land wells, a cellar is usually constructed. The cellar accommodates the wellhead, and allows the tree to be positioned with the base at ground level. The first casing string, conductor, is piled in to the depth of refusal. Alternatively, a post hole may be drilled and the casing run and cemented in place. On offshore platforms, the surface casing is normally run through guide slots that are part of the jacket structure and piled in to the seabed. In subsea wells, the casing is normally drilled and cemented into place. If the seabed is soft, it may be jetted into place.

With the conductor in place, the well is drilled to a depth where the surface casing can be run. Whilst the surface casing hole is being drilled, a diverter is in place on the surface to deal with shallow gas blowout. The surface casing is normally the first pipe that will support the drilling BOP, and the hole should be deep enough to ensure that formation fracture pressure is high enough to withstand kick pressure as the next hole size is being drilled. In most cases, the surface casing is cemented to the surface (or seabed). Once the surface casing has been run, successfully cemented, and tested, the drilling BOP will be installed.

Intermediate casing is used to deepen the wells where kick tolerance or unstable formation mean that it is not possible, or desirable, to drill from the surface casing to the reservoir in one continuous hole section. In very deep or difficult-to-drill wells, more than one intermediate casing string may be required. Cementing of intermediate strings will depend on a number of factors. For example, government regulation may require all or some formations to be isolated, particularly before abandonment. This may mean having to put the top of the cement above the shoe of the previous (surface) casing string. In other cases, a deliberate decision might be made not to cement above the shoe. Placing the top of the cement below a shoe might allow the casing to be cut and pulled, should a side track be needed. It may also allow annulus pressure to leak off into the formation at the exposed shoe. This is deliberate, and can help reduce annulus pressure build-up because of rising temperature as a well begins to produce (Fig. 2.1).

Some wells are drilled from the surface (or intermediate) casing shoe all the way through the reservoir to total depth (TD). Production casing is run across the producing formation and cemented in place. The casing is then perforated to allow hydrocarbons to enter the wellbore, where they will be produced through the completion.

In many wells, the production casing is set (and cemented) above the producing interval. Once the casing has been tested, the shoe of the

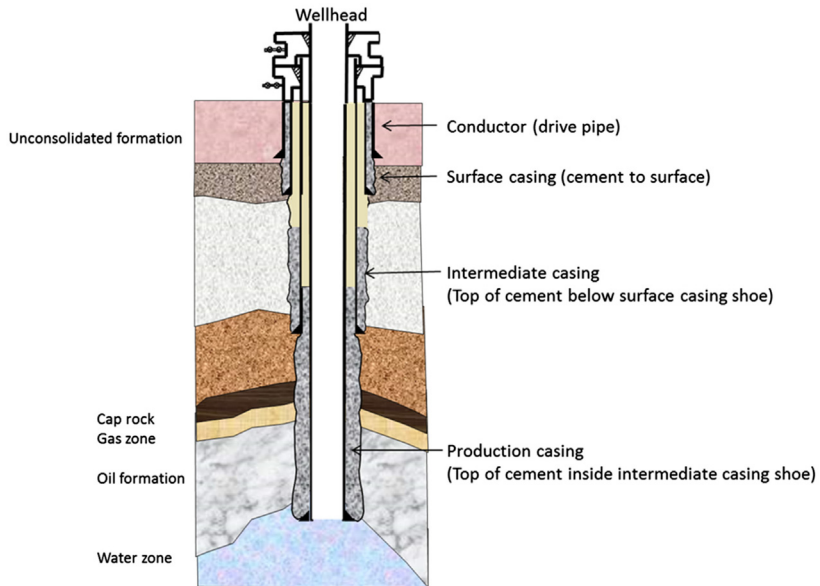


Figure 2.1 Casing and cementing the wellbore.

production casing is drilled out, the reservoir drilled, and the well completed. The type of completion will depend on the nature of the reservoir, and the different types of reservoir completion are discussed next.



2.2 TYPES OF COMPLETION

A completion is the final phase of the well construction process, and forms the interface between the reservoir and surface production facilities.

Wells can be producers or injectors, and completions may have to be able to produce (or inject) hydrocarbon liquids and gas, water, steam, and a range of stimulation chemicals. In addition, completions may have to withstand the corrosive effects of nonhydrocarbon products such as CO_2 and H_2S . Some completions will combine functions, e.g., annulus water injection and through tubing production. Over the life of the well, conditions can change and some wells may start life as production wells, and be converted to injection as conditions in the reservoir change.

Completion engineers often differentiate between reservoir completions (alternatively called the lower or sand-face completion) and the upper completion. The reservoir completion forms the connection between the reservoir and the wellbore. The upper completion is the conduit between the reservoir completion and the surface facilities.

2.2.1 The reservoir completion (sand-face or lower completion)

A good completion engineer will focus much of their attention on the lower completion design, since it is the efficiency of the lower completion that has the biggest influence on well productivity. Lower completion design will fall into one of only five categories. Within those five options there are a huge number of variables. Getting it wrong will be costly.

The main design decisions affecting the choice of reservoir completion are:

- Well trajectory and inclination
- Open hole vs cased hole
- Sand control requirements
- Stimulation requirements (propped frac or acid stimulation)
- Reservoir conformance — single or multi-layered reservoir and isolation between zones.

The five categories of reservoir completion are:

1. Open hole barefoot. This type of completion is simply a hole drilled through the hydrocarbon producing zone.
2. Open hole with slotted or predrilled liner.
3. Cased and perforated. A cemented liner or casing string is run across the reservoir and cemented in place. The casing is then perforated to provide a communication path between the reservoir and the wellbore.
4. Sand control completions in open hole. Sand control completions are required where sand production is anticipated. Sand production has detrimental effects on completions if not controlled. Open hole sand control completions include stand-alone screens and gravel packed screens.
5. Sand control screens in cased hole. Sand control screens can be run in to existing cased and perforated wells, or newly completed cased wells, and include screen-only completions, gravel pack, or frac pack. Expandable sand screens can and have been run in both open hole and cased hole (Fig. 2.2).

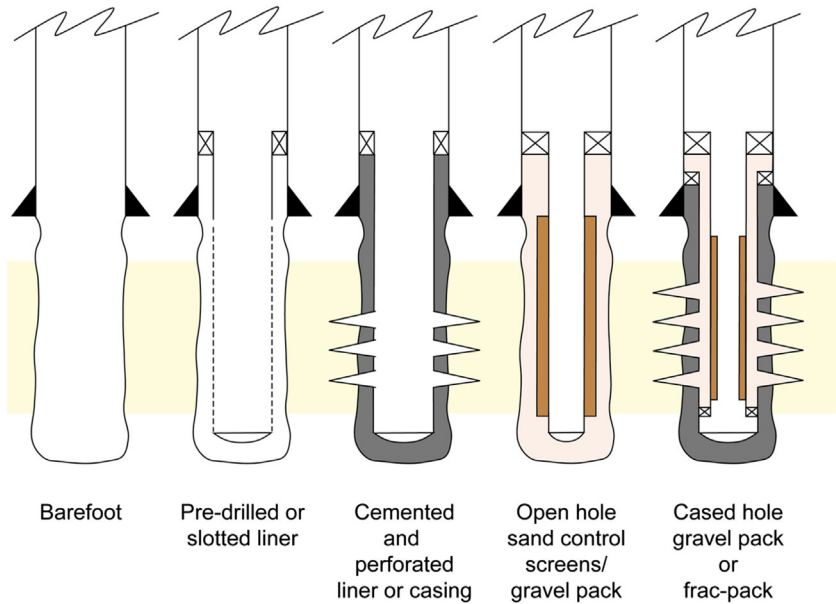


Figure 2.2 The five reservoir completion methods.

There are advantages and disadvantages associated with each of the five main completion types, and these are worth exploring further.

2.2.2 Open hole (barefoot) completions

Open hole completions are fairly common in homogeneous carbonate (limestone and dolomite) reservoirs, where rock strength is generally high and drilling induced formation damage can easily be removed with acid ([Fig. 2.3](#)). They are less common in sandstone formations. The main advantages and disadvantages of open hole (barefoot) completions are summarized in [Table 2.1](#).

2.2.3 Open hole completions with predrilled and slotted liner

Predrilled and slotted liners are generally used where there is a risk of gross hole collapse. Using a predrilled or slotted liner also allows isolation between production zones if the liner is run in combination with External Casing Packers (ECP), or more commonly nowadays, swell packers. With a liner in place the deployment of intervention tools becomes simpler. For example, bridge plugs can be set in the blank pipe inside a Swell packer to isolate water production from a deeper zone.

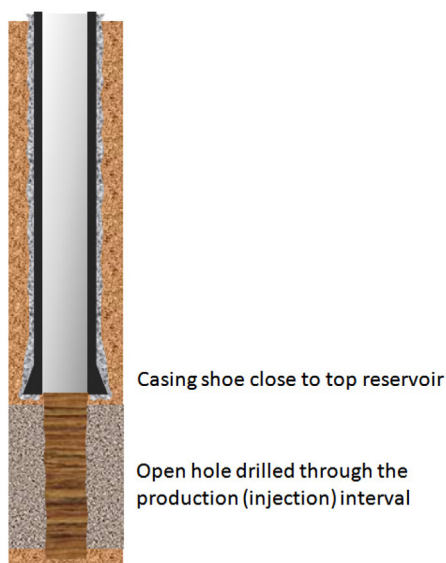


Figure 2.3 Open hole barefoot completion.

Table 2.1 Open hole completions summary

Advantages	Disadvantages
No casing and perforating costs	Water or gas entry difficult to control
Saves rig time	Not easy to selectively stimulate different parts of the producing interval
Easy to deepen or side-track if desired	Open hole section subject to collapse—may require frequent clean-out
Can be converted to screen or liner completion at a later date	Only applicable to well consolidated formation
Large inflow area—high rate production	Drilling damage can be difficult to remove and can lead to reduced inflow

Predrilled and slotted liners are not normally used for sand control; it is difficult to cut slots fine enough to hold back the sand pack. Where very small slots are used (laser cut) the flow area is small, and they become susceptible to plugging. Pre drilled liners are generally preferred because of better inflow performance; they are also cheaper to manufacture.

Predrilled and slotted liners can be deployed with washpipe if there are concerns about reaching bottom. In many cases, the liners are run with mud still in the hole. This reduces the risk of surge and swab induced kicks or losses (Figs. 2.4 and 2.5; Table 2.2).



Figure 2.4 Slotted liner. Source: *Image courtesy of Euroslot.*

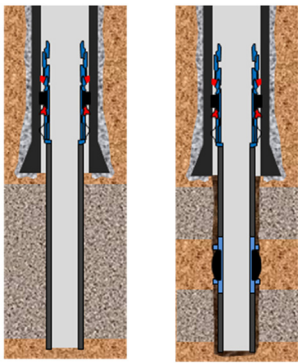


Figure 2.5 Predrilled or slotted liner completion. Left: Single zone. Right: Multiple zone with ECP or swell packer zonal isolation.

Table 2.2 Predrilled and slotted liner completions summary

Advantages	Disadvantages
Relatively cheap—elimination of cementing costs (and time)	Water or gas entry difficult to control unless external casing packers or “Swell” packers are used
Elimination of perforating costs	Not easy to selectively stimulate different parts of the producing interval unless ECP’s or swell packers are used
Slots can be sized to give a degree of sand control in coarse sand formations	Mud filter cake very difficult to remove and can leave residual damage
Some degree of borehole stability achieved	Reduction in diameter over producing interval Some production logs difficult to interpret—at worst they become meaningless

2.2.4 Cased and perforated completion

Cased and perforated completions are used extensively in a variety of reservoir types. Wells can be completed with either a cemented production liner or cemented production casing across the reservoir. Cased and perforated completions have many advantages over open hole completions. These are summarized in [Table 2.3](#): Cased and perforated completions ([Fig. 2.6](#)).

2.2.5 Sand control completions

Although approximately 60% of the world's hydrocarbon production comes from carbonate reservoirs, almost 90% of the wells are in sandstone formations.¹ About 30% of these sandstone formations are unconsolidated, and weak enough to be at risk of sand production.² The type of sand control completion used is dependent upon the consolidation of the

Table 2.3 Cased and perforated completions

Advantages	Disadvantages
Selectivity in production/injection	Perforating costs can be significant—particularly for long horizontal wells
Ability to shut off water, gas, or sand production through relatively straightforward interventions—plugs, straddles, cement squeeze	Some risk of productivity impairment from “completion skins” depending on how the well has been drilled and completed
Good productivity—drilling related formation damage can be by-passed	Adequate cement bond may be difficult to obtain in wells with long high angle liners
Zones can be added at a later date.	
Poorly producing zones can be re-perforated to by-pass damage	
Suitable for fracturing/stimulation, particularly where fracture containment or multiple fracturing is required	
Reduced sanding through careful perforating design	
Ease of application of chemical treatments—diversion	
Easily adapted for “smart” completions—single string multiple zone with multiple packers and inflow valves	

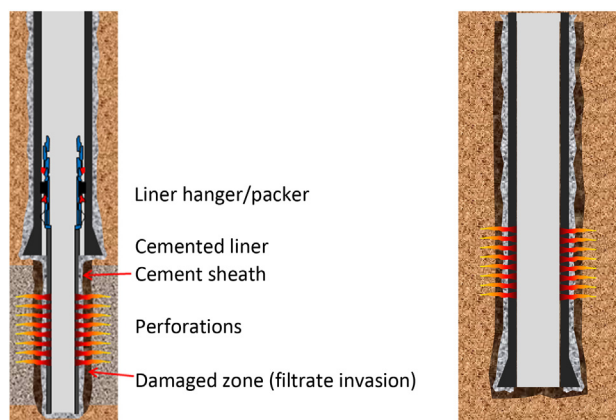


Figure 2.6 Cased and perforated completions. Left: Perforated production liner completion. Right: Perforated production casing completion.

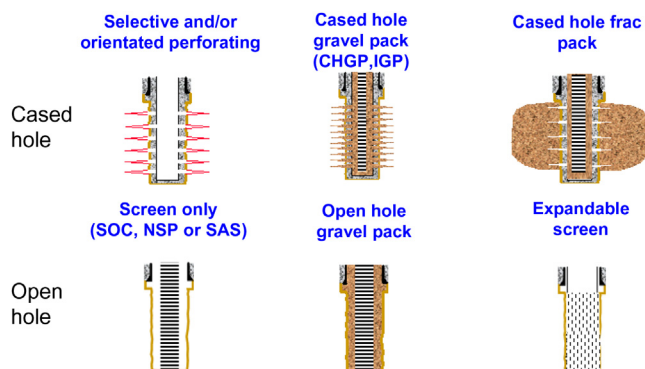


Figure 2.7 Sand control completion types.

formation (rock strength), homogeneity within the reservoir, and the quality of the sand pack (grain size distribution). Good quality well sorted coarse grained sandstone with a small percentage of fines can be completed with screen-only completions. As the quality of the sand deteriorates, more complex solutions are needed; open hole gravel pack or cased hole frac pack. The main sand control completion options are shown in [Fig. 2.7](#).

A screen type sand control completion is run in two parts. Firstly, the screens are run into the open hole and hung off using a gravel-pack packer, an operation analogous to running a liner. If the well is to be gravel packed, it would take place at this point—before the running string is pulled from the well. With the lower completion in place, the upper

Table 2.4 Sand control completions summary

Advantages	Disadvantages
Sand production eliminated or significantly reduced	Costly Difficult to install Risk of high skin and increasing skin over time with many gravel pack/frac pack completions Poor record of reliability—mean time to failure for screen-only completions is 7 years Zonal isolation, water and gas shut-off difficult in open hole sand control completions Screen failure normally means extensive length and costly workover. Side track normally needed

completion (production tubing) can be installed. Screens can be run with (highly conditioned) mud still on the well, and the mud flowed back through the screens and gravel (if used). Some operators displace the mud from the well, replacing it with clear filtered brine, before the screens are run. There is a risk from losses unless a properly formulated lost circulation material is used. The LCM must be sized to prevent losses, but still enable flow-back through the screens (and gravel). There are well control problems particular to the running of screens, and these are described fully in Chapter 7, Well Kill, ([Table 2.4](#)).

2.2.6 The upper completion

The upper completion is the conduit taking the hydrocarbons from the lower (reservoir) completion to the surface. A number factors will influence the configuration of the upper completion. These will include:

- Pressure isolation requirements between reservoir and surface—is a packer required?
- Integrity requirements—Surface Controlled Sub Surface Safety Valve (SC-SSSV) and Annulus Safety Valve (ASV).
- Reservoir management requirements—single string multizone completion or dual string completion.
- Tubing size (driven by production potential and expected decline).
- Artificial lift requirement.
- Life of field concerns.
- Intervention requirements.

Some of the more common upper completion configurations are illustrated and discussed in the next few pages.

2.2.7 No tubing: Flow through production casing

Tubingless completions, if completion is indeed the right word, are still in use in many places. They are popular first and foremost because they are cheap and simple. They are primarily used on land locations where conditions are benign. There are clear disadvantages to this type of “completion.”

- Well integrity—The production casing and wellhead is exposed to reservoir pressure. There are no barriers in the event of a loss of wellhead integrity. Quite simply, they are not safe.
- The casing is exposed to wellbore products, and is at risk of erosion and corrosion leading to well integrity problems.
- Flow stability concerns in larger casing sizes.
- Well kill (when necessary) is complicated by the absence of a tubing string. Bullheading is the only option, unless coil tubing or a snubbing unit is used to create a circulation path.

From a well integrity and well control standpoint, these wells fall far short of the ideal and are not allowed by some regulatory authorities—particularly if the well is capable of flow to surface without reliance on artificial lift ([Fig. 2.8](#)).

2.2.8 Tubing only completion: no production packer

Running a string of completion tubing gives number of distinct advantages over the tubingless configuration.

- Although the casing is still exposed to hydrocarbons and other wellbore products, the annulus between the tubing and casing is static. The risk of erosion is virtually eliminated, and whilst corrosion is still possible it will be at a much reduced rate since the fluids in the annulus are static (not being replenished by flow), unless the well is deliberately produced through the annulus.
- Worn or corroded tubing can be recovered and replaced—whereas the repair of corroded casing would be more complex and costly, perhaps requiring a side track or even abandonment.
- Tubing can be correctly sized to stabilize flow.
- Well kill is simplified—reverse circulation or conventional circulation now becomes possible.

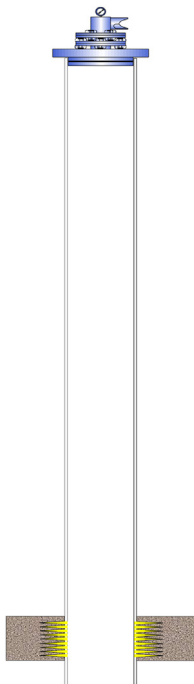


Figure 2.8 A tubingless completion.

The tubing only (packerless) completion is the most common method of upper completion in use today, since it includes almost all rod pump wells (Beam Pump and Progressive Cavity Pump) as well as many wells equipped with Electrical Submersible Pumps (ESP). However, if the well is capable of flowing to surface naturally, i.e., with the artificial lift pump switched off, then integrity remains a concern. The wellhead and casing will be exposed to reservoir pressure unless a packer is installed (Fig. 2.9).

2.2.9 Single production tubing string with production packer (reverse taper configuration)

Adding a production packer to the string of tubing removes many of the integrity concerns associated with the packerless completion.

- The casing above the packer is isolated from wellbore fluids, therefore corrosion is much less likely to occur.
- The wellhead and casing (above the packer) are no longer exposed to reservoir pressure.

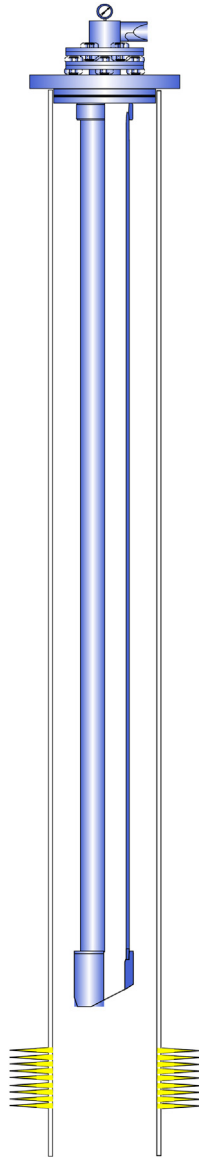


Figure 2.9 Packerless (tubing only) completion.

- On the annulus side there are two mechanical barriers between the reservoir and surface—packer and tubing hanger. These will be integrity tested at the time of installation, and form part of a well integrity barrier system.

- On the tubing side, integrity is provided by closure of the Christmas tree valves. An additional barrier can be provided in the form of a SC-SSSV.
Installing a packer in the well brings with it additional complications.
- Annulus pressure build-up (APB) caused by fluid expansion must be anticipated and controlled by regular monitoring and bleed off. Failure to do so could result in tubing collapse or casing burst.
- The annulus usually becomes an integral part of the well integrity envelope (the secondary barrier), and therefore annulus integrity must be monitored and maintained. Loss of integrity will almost certainly require a well intervention.
- With a packer in place, a circulation path is no longer available. Circulation, for well kill or any other purpose, can only take place if a circulation path is created. This must be close to the packer (for maximum effect). Some completions will include a circulation device, such as a sliding sleeve or side pocket mandrel. In the absence of circulation equipment, the only method of creating a circulation path is to perforate (punch) the production tubing above the packer.
- Workover (re-completion) becomes more complex. If the tubing is connected (anchored) to the packer it must be released before it can be recovered. If a dynamic (moving) seal assembly is stabbed into the packer, then the completion below the packer remains in place unless the packer is recovered separately. Packer replacement can be problematic, if permanent packers have to be removed.
- The tubing will be subjected to much higher loads (axial, burst collapse, and tri-axial). Detailed tubing stress analysis will need to be carried out to confirm the design (Fig. 2.10).

2.2.10 Single production tubing string with production packer (mono-bore configuration)

A mono-bore completion is generally defined as one where the internal diameter of the production tubing is the same or larger than the internal diameter of the production liner.

A completion configured in this way has some distinct advantages.

- Water and gas shut-off interventions can be easily accomplished using full bore (liner ID) mechanical bridge plugs, packers, and straddles.
- There is no need to kill the well and recover the production tubing to gain access to the reservoir.
- Prior to a workover (re-completion), full size mechanical bridge plugs can be run through the upper completion and set in the liner, above

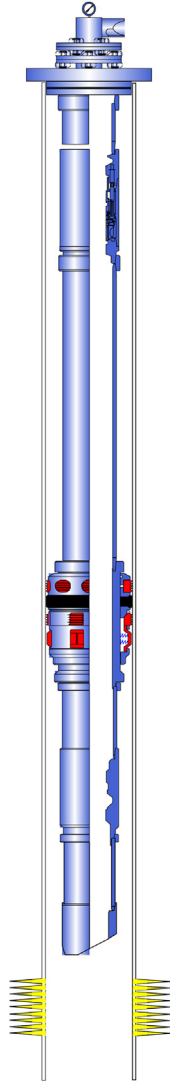


Figure 2.10 Single string with packer.

the reservoir. Having a mechanical barrier in the well during a work-over reduces the risk of fluid loss or a kick, as well as preventing potentially damaging kill fluid from contacting the formation.

- Interventions to remove oilfield scale from the liner can be performed using a full size scale mill. There is no requirement to use more troublesome underreamers.

There are some disadvantages with the mono-bore design. Frictional pressure drop, particularly in long small ID liners, can be significant. In addition, obtaining a good cement bond in high angle wells with long liners can be problematic (Fig. 2.11).

2.2.11 Single string production tubing completed across multiple reservoir zones

Many fields have a number of separate reservoir compartments stacked vertically one above another. Reservoirs of this type can be exploited using a single string, multiple zone, completion. Production zones are isolated from one another using production packers, and there will normally be an additional (top) packer isolating the production annulus from produced fluids.

Fig. 2.12 illustrates a single string multizone completion. In the example illustrated, a well is completed across three separate zones. A configuration like this allows simultaneous production from any combination of zones, or would allow a single zone to be produced on its own. In the example shown, production from the lower zone enters the tubing through the tail-pipe below the lower packer. Flow from this zone could be isolated by installing a wireline set plug (mechanical barrier) in a nipple profile below the packer. Production from the middle and upper zone would be controlled by opening or closing a communication port (sliding sleeve) located between the packers. In most wells the communication port is manipulated using a slickline deployed mechanical shifting tool. In wells where access is difficult or expensive, for example subsea wells, communications ports are hydraulically operated from the surface (“intelligent” or “smart” well technology).

One advantage of this design is the number of different reservoir layers that can be completed using a single wellbore and a single completion string—in theory there is no limit. Fields are in production where eight or nine zones are accessed through a single completion string. There are some obvious disadvantages to completing in this way. Intervention access is very limited at the zones that are behind pipe. Pressure contrast between zones can lead to problems with cross-flow. This can compromise well productivity, create problems during intervention, and make well kill very problematic. There will be additional well control concerns when running the completion, since the production casing will have to be perforated before the completion is run.

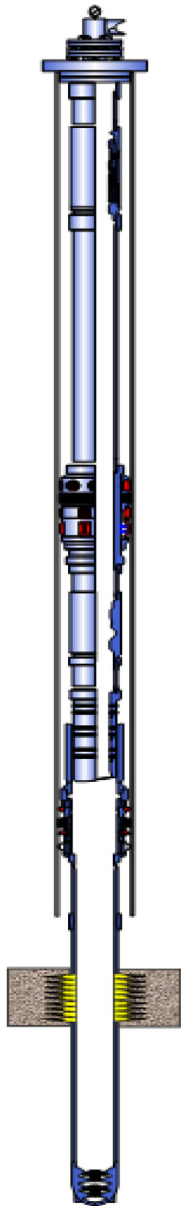


Figure 2.11 Mono-bore completion.

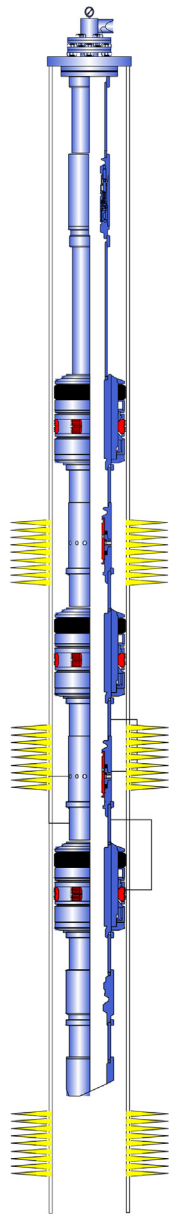


Figure 2.12 Single string multizone completion.

2.2.12 Multilateral Wells

A multilateral well is a well with one or more branches coming off the main bore. Because of the difficulty in drilling and completing multilateral wells, they are still relatively rare. The main reason for drilling a multilateral well is to increase reservoir exposure. For example, in the Troll Field, Statoil have drilled a well that uses five branches to expose 14,200 m (46,590 ft) of producing formation. Whilst there are clear benefits to having such extensive reservoir exposure from a single wellbore, the drawbacks are equally as obvious. Multilateral wells are complex; difficult to drill and complete. Any requirement to control production from a lateral means complex completions. The ability to carry out mechanical interventions is normally limited to the main bore, unless very complex junctions are installed. There are also additional well control problems associated with some multilateral wells—for example, a bullhead may be the only viable option in some wells, since it is usually not possible to circulate below the junction (Fig. 2.13).

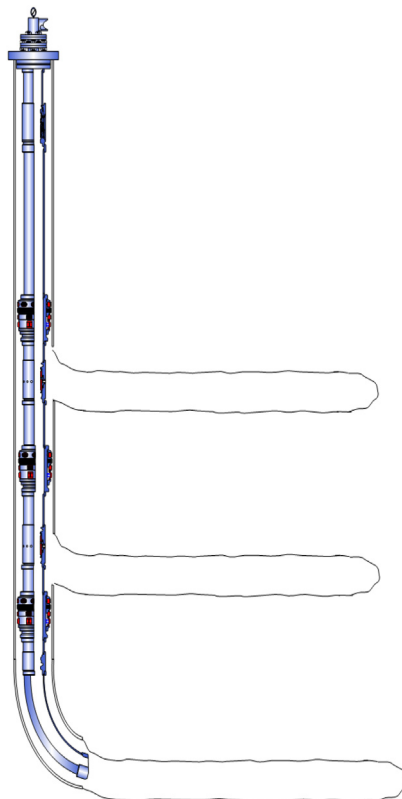


Figure 2.13 Multilateral completion.

In 1997 a joint industry taskforce, the Technical Advancement for Multilaterals (TAML) established a six tier classification for isolation at the junction. The six junction categories are illustrated, each with a brief description (Fig. 2.14).

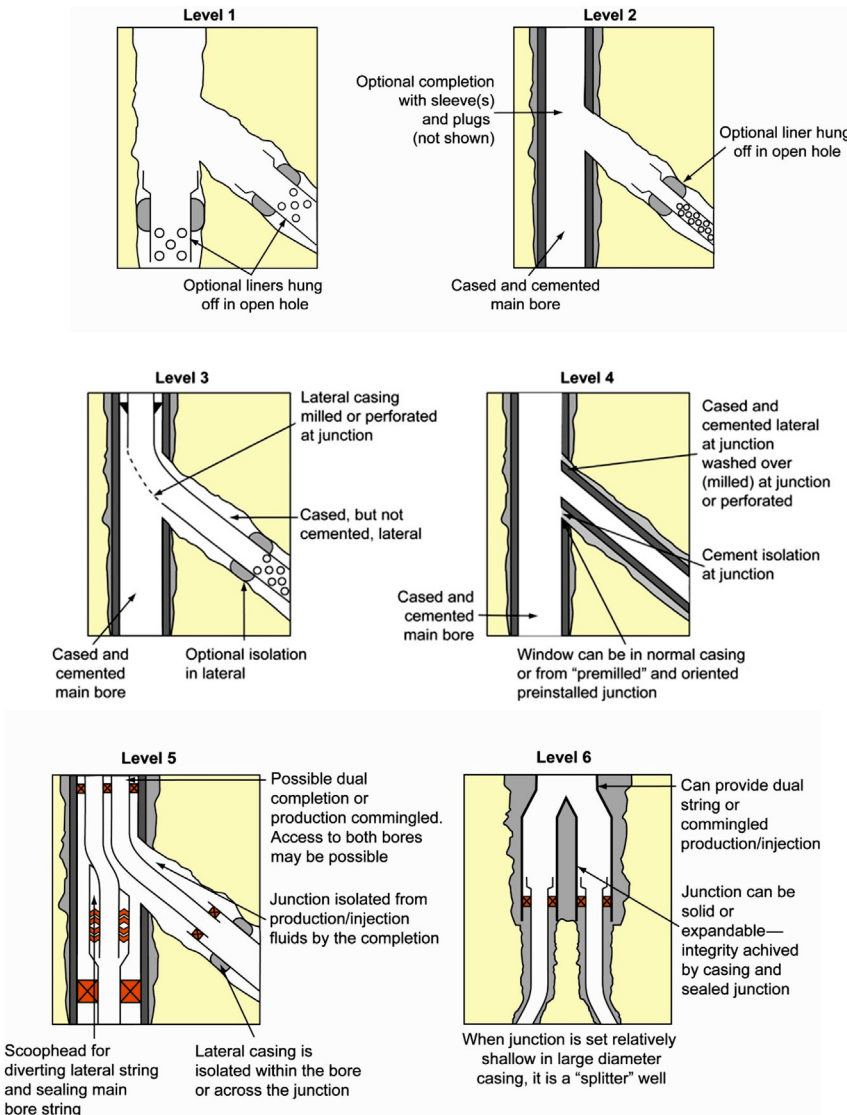


Figure 2.14 Technical Advancement for Multilateral (TAML) junction classification.



2.3 SUMMARY

Anyone with responsibility for well control or well integrity during completion, workover, or intervention operations must have at least a basic understanding of the well construction process, as well as an in-depth knowledge of completion types and completion architecture. Knowing how the well works as an integrated system will aid identification of potential leak paths, and how best to respond to a well control incident should one occur.

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